# ITSAC Supplementary Information

A Ground–Low-Altitude Mobility Manager for Service Reconfiguration under Disruptions

## S1 Detailed experimental outcomes

The tables report service outcomes from the revised numerical analysis. Means use the original fixed scenario weights, and intervals use seed blocks. The B0 D60 results include the closed-loop duplicate-command correction. The restored branch retains the historical analysis and source files. The new eight-page manuscript summarizes these archived experiments; no new simulation or inference cohort is introduced.

Table S1: E1 service outcomes: mean [95% interval].

| Site | Policy | n | Completion (s) | Deadline violations (%) | Damaged services/run |
|---|---|---|---|---|---|
| A | B0 | 240 | 187.67 [187.67, 187.67] | 33.33 [33.33, 33.33] | 0.000 [0.000, 0.000] |
| A | B1 | 240 | 160.78 [159.13, 162.42] | 25.00 [25.00, 25.00] | 0.500 [0.500, 0.500] |
| A | B2 | 240 | 147.62 [147.20, 148.04] | 16.67 [16.67, 16.67] | 0.058 [0.040, 0.077] |
| A | B4b | 240 | 146.63 [145.88, 147.38] | 16.67 [16.67, 16.67] | 0.133 [0.114, 0.153] |
| B | B0 | 240 | 262.67 [262.67, 262.67] | 50.00 [50.00, 50.00] | 0.000 [0.000, 0.000] |
| B | B1 | 240 | 47.70 [46.49, 48.91] | 0.00 [0.00, 0.00] | 0.500 [0.500, 0.500] |
| B | B2 | 240 | 47.70 [46.49, 48.91] | 0.00 [0.00, 0.00] | 0.500 [0.500, 0.500] |
| B | B4b | 240 | 49.10 [47.03, 51.17] | 0.00 [0.00, 0.00] | 0.487 [0.473, 0.502] |
| C | B0 | 240 | 323.67 [323.67, 323.67] | 83.33 [83.33, 83.33] | 0.000 [0.000, 0.000] |
| C | B1 | 240 | 168.28 [165.32, 171.23] | 10.00 [4.12, 15.88] | 0.500 [0.500, 0.500] |
| C | B2 | 240 | 168.28 [165.32, 171.23] | 10.00 [4.12, 15.88] | 0.500 [0.500, 0.500] |
| C | B4b | 240 | 168.65 [165.58, 171.72] | 10.00 [4.12, 15.88] | 0.500 [0.500, 0.500] |

Completion, damage and deadline-rate intervals use 20 seed blocks, each averaging the same 12 scenarios. A zero-width interval means that this fixed-panel average did not vary across seeds; it does not establish a universal rate. Each site–policy cell contains 240 runs.

Table S2: E2 all-run transport outcomes and conditional path restoration.

| Site | Policy | All runs | Completion (s), mean [95% CI] | Deadline violations (%) | Recovered/affected | Restoration (%) |
|---|---|---|---|---|---|---|
| A | B0 | 320 | 182.00 [182.00, 182.00] | 100.00 | N/A | N/A |
| A | B1 | 320 | 162.00 [162.00, 162.00] | 37.50 | 200/200 | 100.00 |
| A | B2 | 320 | 162.00 [162.00, 162.00] | 37.50 | 200/200 | 100.00 |
| A | B4b | 320 | 166.53 [164.41, 168.64] | 39.06 | 191/191 | 100.00 |
| B | B0 | 320 | 241.00 [241.00, 241.00] | 100.00 | N/A | N/A |
| B | B1 | 320 | 153.88 [153.88, 153.88] | 37.50 | 200/200 | 100.00 |
| B | B2 | 320 | 153.88 [153.88, 153.88] | 37.50 | 200/200 | 100.00 |
| B | B4b | 320 | 154.05 [153.94, 154.16] | 37.50 | 199/199 | 100.00 |
| C | B0 | 320 | 310.00 [310.00, 310.00] | 100.00 | N/A | N/A |
| C | B1 | 320 | 278.00 [278.00, 278.00] | 62.50 | 200/200 | 100.00 |
| C | B2 | 320 | 278.00 [278.00, 278.00] | 62.50 | 200/200 | 100.00 |
| C | B4b | 320 | 278.92 [277.79, 280.06] | 62.81 | 201/201 | 100.00 |

All-run outcomes use 320 runs per site and policy. Recovery is conditional on the primary air chain actually being affected; it is N/A for B0. Restoration is reported as observed recovered/affected counts. We retain dependence within the fixed panel and do not apply a Wilson interval that assumes independent runs.

Table S3: Mean E3 system weighted loss by disturbance/task level.

| Site | Level | Runs/policy | B0 | B1 | B2 | B4b |
|---|---|---|---|---|---|---|
| A | L1 | 40 | 240.000 | 0.000 | 0.000 | 6.000 |
| A | L2 | 80 | 240.000 | 240.000 | 240.000 | 240.000 |
| A | L3 | 80 | 240.000 | 0.000 | 0.000 | 1.688 |
| A | L4 | 40 | 240.000 | 120.000 | 120.000 | 123.375 |
| B | L1 | 40 | 240.000 | 0.000 | 0.000 | 0.000 |
| B | L2 | 80 | 240.000 | 120.000 | 120.000 | 120.000 |
| B | L3 | 80 | 375.000 | 135.000 | 135.000 | 135.000 |
| B | L4 | 40 | 375.000 | 255.000 | 255.000 | 255.000 |
| C | L1 | 40 | 240.000 | 240.000 | 240.000 | 240.000 |
| C | L2 | 80 | 240.000 | 240.000 | 240.000 | 240.000 |
| C | L3 | 80 | 375.000 | 135.000 | 135.000 | 135.000 |
| C | L4 | 40 | 375.000 | 375.000 | 375.000 | 375.000 |

L1/L4 have 40 runs and L2/L3 have 80 per policy. Levels change disturbance structure and demand. SWL uses flat configured penalties and priority weights, rather than monetary costs.

Table S4: E4A observation scheduling, completion and B4b input demand.

| Interval (s) | B0 completion (s) | B1 completion (s) | B2 completion (s) | B4b completion (s) | B4b late (%) | B4b decisions/run | B4b prompt tokens/run |
|---|---|---|---|---|---|---|---|
| 10 | 192.0 | 140.0 | 140.0 | 140.0 | 0 | 62 | 172849.0 |
| 30 | 212.0 | 150.0 | 150.0 | 150.0 | 0 | 22 | 62307.6 |
| 60 | 253.0 | 180.0 | 180.0 | 180.0 | 0 | 12 | 35017.4 |
| 120 | 253.0 | 240.0 | 240.0 | 240.0 | 100 | 7 | 21348.2 |
| 300 | 482.0 | 521.0 | 482.0 | 482.0 | 100 | 4 | 11707.6 |

Each policy–arm cell contains 20 runs. Decision counts cover the complete scheduled polling horizon, including post-resolution observations. Token means retain recorded backend outcomes.

Table S5: E4B injected execution delay and mean completion duration.

| Delay (s) | B0 (s) | B1 (s) | B2 (s) | B4b (s) | B4b late (%) |
|---|---|---|---|---|---|
| 0 | 182.0 | 120.0 | 120.0 | 120.0 | 0 |
| 1 | 183.0 | 121.0 | 121.0 | 121.0 | 0 |
| 5 | 187.0 | 125.0 | 125.0 | 125.0 | 0 |
| 10 | 192.0 | 130.0 | 130.0 | 130.0 | 0 |
| 20 | 202.0 | 140.0 | 140.0 | 140.0 | 0 |
| 30 | 212.0 | 150.0 | 150.0 | 150.0 | 0 |
| 60 | 242.0 | 341.0 | 302.0 | 302.0 | 100 |

Each policy–arm cell contains 20 runs. Waiting times are controlled simulation interventions, not measured inference latency.

Table S6: E4C B4b input burden with an unchanged core fleet.

| Total aircraft records | Core aircraft | Completion (s) | On time (%) | Logged calls (20 runs) | Prompt tokens/logged call |
|---|---|---|---|---|---|
| 5 | 4 | 121.5 | 100 | 41 | 3982.1 |
| 10 | 4 | 120.0 | 100 | 40 | 5920.6 |
| 20 | 4 | 120.0 | 100 | 40 | 9588.7 |
| 30 | 4 | 120.0 | 100 | 40 | 13256.7 |
| 50 | 4 | 120.0 | 100 | 40 | 20593.5 |

Each arm contains 20 B4b runs. Added records represent unavailable and non-commandable aircraft. Token means include retained error calls with zero recorded tokens; active fleet size stays four.

Table S7: Aggregate E3 outcomes across the 12-scenario panel.

| Site | Policy | n | Completion (s), mean [95% CI] | Deadline violations (%) | SWL, mean [95% CI] |
|---|---|---|---|---|---|
| A | B0 | 240 | 182.00 [182.00, 182.00] | 100.00 | 240.000 [240.000, 240.000] |
| A | B1 | 240 | 187.25 [187.25, 187.25] | 41.67 | 100.000 [100.000, 100.000] |
| A | B2 | 240 | 187.25 [187.25, 187.25] | 41.67 | 100.000 [100.000, 100.000] |
| A | B4b | 240 | 188.80 [187.18, 190.41] | 42.08 | 102.125 [99.578, 104.672] |
| B | B0 | 240 | 241.00 [241.00, 241.00] | 100.00 | 307.500 [307.500, 307.500] |
| B | B1 | 240 | 146.83 [146.83, 146.83] | 25.00 | 127.500 [127.500, 127.500] |
| B | B2 | 240 | 146.83 [146.83, 146.83] | 25.00 | 127.500 [127.500, 127.500] |
| B | B4b | 240 | 147.05 [146.78, 147.32] | 25.00 | 127.500 [127.500, 127.500] |
| C | B0 | 240 | 310.00 [310.00, 310.00] | 100.00 | 307.500 [307.500, 307.500] |
| C | B1 | 240 | 322.67 [322.67, 322.67] | 66.67 | 227.500 [227.500, 227.500] |
| C | B2 | 240 | 322.67 [322.67, 322.67] | 66.67 | 227.500 [227.500, 227.500] |
| C | B4b | 240 | 323.82 [322.97, 324.68] | 66.67 | 227.500 [227.500, 227.500] |

Scenario–seed runs receive equal weight. L2/L3 therefore contribute twice as many runs as L1/L4; equal-level weighting would define a different summary.

# S2 Interface and implementation details

## S2.1 Principal notation

Table S8: Principal notation.

| Symbol | Meaning | Unit |
|---|---|---|
| $t, \tau(t), t_e, H$ | Decision time, observation timestamp, scheduled execution and horizon | s |
| $S_t, \widehat{S}_t$ | Current joint state and observed state | — |
| $r_m, d_m, c_m$ | Mission release, deadline and completion | s |
| $\mathcal{I}_t, \mathcal{C}_{m,t}$ | Candidate information and legal subset | — |
| $\widehat{\eta}_c, \delta_{\text{exec}}, \sigma_m$ | Remaining completion estimate, execution delay, estimated temporal margin | s |
| $a^{\text{prop}}, u, Q, \Gamma$ | Proposal, executed intervention, current-version indicator, current-state checker | — |
| $J_t(c)$ | Specified one-step heuristic score | Configured seconds-equivalent units |
| $A_i, R_i$ | Affected-chain and recovered-path indicators | Binary |
| $T_m, V_m, L_H$ | Completion duration, deadline violation, weighted service loss | s; binary; loss units |

## S2.2 Structured action vocabulary

Table S9: Structured action vocabulary.

| Action | Meaning |
|---|---|
| DISPATCH | Assign an available compatible aircraft |
| REASSIGN | Transfer an eligible occupied aircraft to another mission |
| REROUTE | Change an ongoing operation's route |
| DIVERT | Change an air operation's destination or continuation |
| DELAY | Postpone an operation |
| CANCEL | Cancel a specified operation |
| RESERVE | Reserve a resource for a mission |
| RETURN | Request return by an eligible aircraft |
| LAND | Request landing under modeled constraints |
| GROUND_FALLBACK | Establish ground service for an eligible mission |
| NO_ACTION | Make no operational change at the decision epoch |
| ESCALATE | Report a need beyond the current action choice |

Each scenario uses only part of this vocabulary. Including an action type in the contract does not mean that the experiments validate every possible use of it. The candidate interface records legal flags and rejection reasons, and the checker applies conditions specific to each action.

## S2.3 Policy inputs and parameters

The complete template is supplied as `prompts/manager_v2.txt`, together with the input renderer and action schema. Both B4a and B4b receive the full global snapshot without its `candidates` field. B4b receives that field in a separate JSON section; B4a does not. Both retain raw resource state, mission assignments, priorities, positions, deadlines, ground ETA, prompt rules and the common checker. Only B4b receives derived air ETA, completion predictions, per-candidate legality/rejection reasons and the explicit mapping to preempted missions. The ablation therefore tests explicit candidate-information support, including both derived information and its presentation.

Table S10: Input-field comparison for the E1 ablation.

| Field | B4a | B4b |
|---|---|---|
| Raw aircraft and mission state; ground ETA | Included | Included |
| Rules, objectives, output schema, checker | Same | Same |
| Derived candidate air ETA and completion time | Omitted | Included |
| Per-candidate legal flag and rejection reason | Omitted | Included |
| Predicted deadline violation and preemption consequence | Omitted | Included |

Table S11: Key operational and analysis parameters.

| Parameter | Value or implemented rule |
|---|---|
| Core fleet | Two medical UAVs, one logistics UAV, one passenger eVTOL |
| Cruise speed | Medical/logistics 15 m/s; passenger 40 m/s; inspection role 10 m/s (no core inspection aircraft) |
| Nominal altitude | UAV 328 ft; passenger eVTOL 656 ft |
| Arrival radius | 100 m; this can make realized duration shorter than geometric ETA |
| Dispatch overhead | 0 s in candidate estimates |
| Endurance | Initial 1,800 s, reduced by a seeded 0–15%; candidate reserve margin 60 s |
| Busy-aircraft progress | Seeded fraction 0.05–0.40 of its initial route |
| Simulation step / ordinary decision period | 1 s / 30 s while actionable; E4A overrides observation polling |
| Horizon / primary emergency release | 900 s (E3 L3/L4: 1,200 s) / 300 s |
| E1 deadline slack | HIGH 300 s; CRITICAL 180 s |
| E2/E4 primary deadline | 480 s at Site A; site-adapted configurations accompany the data |
| Incumbent default deadline | 1,800 s for initial background missions |
| Ground completion | Execution-time SUMO D1–H1 ETA rounded to seconds, added to issue time |
| LLM decoding | Temperature 0; JSON output; 8,192 output-token budget |
| SWL priority weights | CRITICAL/HIGH/NORMAL/LOW: 4/3/2/1; penalties are task-specific flat values |

B2 weights and the prompt template remain unchanged. The supplied YAML matrices specify mission-specific delay and cancellation penalties, secondary releases, workload assignments and site adaptations. The policy identifier names the configured backend service; the underlying model version has not been verified.

# S3 Statistical and outcome conventions

## S3.1 Fixed-panel estimand and random streams

The runner initializes a fresh `random.Random(seed)` for every scenario and visits aircraft in sorted order. Each aircraft consumes an endurance draw, and each busy aircraft consumes an additional progress draw. SUMO also reuses the same seed. Workload-dependent draw counts change the realizations, but they do not create independently keyed scenario streams. We therefore retain dependence within each seed across the fixed scenario panel and pair policies by scenario and seed. This decision follows the implementation, rather than seed labels or state hashes alone.

For a complete panel, each seed contributes the average of its scenario-specific outcomes or paired differences. Means keep equal scenario–run weights. A t interval uses the standard deviation across 20 seed averages (10 in the E1 ablation). For a conditional endpoint with $N$ retained pairs, let $\bar{d}$ denote the original pair-weighted mean and $K$ the number of represented seeds. The cluster contribution is

$$B_s = \bar{d} + \frac{K}{N} \sum_{i:\, s_i = s} (d_i - \bar{d}).$$

The interval is $\bar{d} \pm t_{.975,K-1}\, \mathrm{sd}(B_s)/\sqrt{K}$. This retains the mean over jointly affected pairs without reweighting seeds that have different exposure counts. Conditional populations can differ by policy. At Site C, B4b has 201 affected runs, whereas the B4b–B2 recovery comparison includes 200 jointly affected pairs.

Continuous tests use these seed contributions, with the original signed-rank fallback for constant differences. Binary tests enumerate policy-label swaps of whole seed blocks and use the total discordant paired difference as the test statistic. This preserves pairing without assuming independent pairs for McNemar inference. Scenarios, endpoints and correction families are not selected by their p-values. Intervals are unadjusted. Holm-adjusted p-values retain the original three E1 ablation, 12 E2 and 12 E3

compound endpoints per site. E1 scenario-specific and E4 single-scenario comparisons keep their original seed units.

The E2 family includes completion, recovery time, replan latency, incumbent damage, candidate reduction, deadline violation, restoration, recovered completion, ground fallback, failure-induced fallback, necessary-fallback correctness and air intervention. E3 compares B4b with B1/B2 at L2–L4 for SWL and CRITICAL+HIGH loss.

Table S12: Complete E2 seed-block comparison family, Site A. Differences are B4b minus B2; binary differences are fractions.

| Site | Endpoint | Pairs | Difference | 95% interval | Holm p |
|---|---|---|---|---|---|
| A | Completion | 320 | 4.525 | [2.412, 6.638] | 0.003069 |
| A | Recovery time | 191 | 3.927 | [0.930, 6.924] | 0.142465 |
| A | Replan latency | 191 | 3.927 | [0.930, 6.924] | 0.142465 |
| A | Damage | 320 | 0.003 | [-0.012, 0.018] | 1.000 |
| A | Candidate reduction | 320 | 0.013 | [0.000, 0.025] | 0.336690 |
| A | Deadline risk | 320 | 0.016 | [0.003, 0.029] | 0.437500 |
| A | Restoration | 191 | 0.000 | [0.000, 0.000] | 1.000 |
| A | Recovered/completed | 191 | 0.000 | [0.000, 0.000] | 1.000 |
| A | Ground fallback | 320 | 0.013 | [0.000, 0.025] | 0.625000 |
| A | Failure fallback | 320 | 0.013 | [0.000, 0.025] | 0.625000 |
| A | Necessary fallback | 320 | -0.016 | [-0.029, -0.003] | 0.437500 |
| A | Air intervention | 320 | -0.028 | [-0.048, -0.008] | 0.142465 |

Table S13: Complete E2 seed-block comparison family, Site B. Differences are B4b minus B2; binary differences are fractions.

| Site | Endpoint | Pairs | Difference | 95% interval | Holm p |
|---|---|---|---|---|---|
| B | Completion | 320 | 0.175 | [0.060, 0.290] | 0.055710 |
| B | Recovery time | 199 | 0.281 | [0.098, 0.465] | 0.055710 |
| B | Replan latency | 199 | 0.281 | [0.098, 0.465] | 0.055710 |
| B | Damage | 320 | 0.000 | [0.000, 0.000] | 1.000 |
| B | Candidate reduction | 320 | 0.000 | [0.000, 0.000] | 1.000 |
| B | Deadline risk | 320 | 0.000 | [0.000, 0.000] | 1.000 |
| B | Restoration | 199 | 0.000 | [0.000, 0.000] | 1.000 |
| B | Recovered/completed | 199 | 0.000 | [0.000, 0.000] | 1.000 |
| B | Ground fallback | 320 | 0.000 | [0.000, 0.000] | 1.000 |
| B | Failure fallback | 320 | 0.000 | [0.000, 0.000] | 1.000 |
| B | Necessary fallback | 320 | -0.003 | [-0.010, 0.003] | 1.000 |
| B | Air intervention | 320 | -0.003 | [-0.010, 0.003] | 1.000 |

Table S14: Complete E2 seed-block comparison family, Site C. Differences are B4b minus B2; binary differences are fractions.

| Site | Endpoint | Pairs | Difference | 95% interval | Holm p |
|---|---|---|---|---|---|
| C | Completion | 320 | 0.922 | [-0.214, 2.057] | 1.000 |
| C | Recovery time | 200 | 0.255 | [-0.138, 0.648] | 1.000 |
| C | Replan latency | 200 | 0.255 | [-0.138, 0.648] | 1.000 |
| C | Damage | 320 | 0.000 | [0.000, 0.000] | 1.000 |
| C | Candidate reduction | 320 | 0.000 | [0.000, 0.000] | 1.000 |
| C | Deadline risk | 320 | 0.003 | [-0.003, 0.010] | 1.000 |
| C | Restoration | 200 | 0.000 | [0.000, 0.000] | 1.000 |
| C | Recovered/completed | 200 | 0.000 | [0.000, 0.000] | 1.000 |
| C | Ground fallback | 320 | 0.000 | [0.000, 0.000] | 1.000 |
| C | Failure fallback | 320 | 0.000 | [0.000, 0.000] | 1.000 |
| C | Necessary fallback | 320 | 0.000 | [0.000, 0.000] | 1.000 |
| C | Air intervention | 320 | 0.000 | [0.000, 0.000] | 1.000 |

Table S15: E3 compound SWL comparisons. CRITICAL+HIGH differences and tests are identical in these cells; both endpoints remain in each 12-test family.

| Site | Level | Base | Pairs | Difference | 95% interval | Holm p |
|---|---|---|---|---|---|---|
| A | L2 | B1 | 80 | 0.000 | [0.000, 0.000] | 1.000 |
| A | L3 | B1 | 80 | 1.688 | [-1.844, 5.219] | 1.000 |
| A | L4 | B1 | 40 | 3.375 | [-3.689, 10.439] | 1.000 |
| A | L2 | B2 | 80 | 0.000 | [0.000, 0.000] | 1.000 |
| A | L3 | B2 | 80 | 1.688 | [-1.844, 5.219] | 1.000 |
| A | L4 | B2 | 40 | 3.375 | [-3.689, 10.439] | 1.000 |
| B | L2 | B1 | 80 | 0.000 | [0.000, 0.000] | 1.000 |
| B | L3 | B1 | 80 | 0.000 | [0.000, 0.000] | 1.000 |
| B | L4 | B1 | 40 | 0.000 | [0.000, 0.000] | 1.000 |
| B | L2 | B2 | 80 | 0.000 | [0.000, 0.000] | 1.000 |
| B | L3 | B2 | 80 | 0.000 | [0.000, 0.000] | 1.000 |
| B | L4 | B2 | 40 | 0.000 | [0.000, 0.000] | 1.000 |
| C | L2 | B1 | 80 | 0.000 | [0.000, 0.000] | 1.000 |
| C | L3 | B1 | 80 | 0.000 | [0.000, 0.000] | 1.000 |
| C | L4 | B1 | 40 | 0.000 | [0.000, 0.000] | 1.000 |
| C | L2 | B2 | 80 | 0.000 | [0.000, 0.000] | 1.000 |
| C | L3 | B2 | 80 | 0.000 | [0.000, 0.000] | 1.000 |
| C | L4 | B2 | 40 | 0.000 | [0.000, 0.000] | 1.000 |

### S3.2 Service-loss audit

In the archived revision, all 2,880 E3 run registries were checked against their recorded horizons: 900 s for L1/L2 and 1,200 s for L3/L4. At each site, 1,440 task instances had deadlines at or before the corresponding horizon, and all were COMPLETED. Across sites, zero overdue tasks remained EN_ROUTE or ASSIGNED. The audit therefore requires no correction to SWL for overdue ongoing tasks. Background missions with 1,800 s deadlines fall outside both horizons; remaining in service does not by itself constitute a failure. Incumbent damage remains a count of damaged final states.

As a complementary E1 measure, we linked recorded preemption times to the incumbent identified in the candidate log and its final interrupted state. At A, exposure from preemption to the horizon for persistently interrupted services averages 35 s per run for B2 and 80 s for B4b (B1: 300 s). This measure is truncated at the horizon; it does not capture the complete cumulative cost of interruption. Mission CSV records omit some incumbent preemption transitions, so a logged interruption duration of zero would be misleading. Initial incumbent deadlines are 1,800 s, beyond the 900 s horizon. This exposure measure therefore supports no claim about incumbent lateness.

## S4 Technical checks and timing sensitivity

### S4.1 Duplicate pending commands and E4B regression

The original B0 D60 pipeline records a GROUND_FALLBACK command at 300 s, due at 360 s. At 360 s, the fault triggers a new decision with the same normalized action and the same 181.763 s ETA field. The original queue replaces the pending command and restarts the 60 s wait. It issues the command at 420 s, and transport completes at 602 s. This extra wait produced the original 302 s completion duration.

The revised queue keeps the original due time when a validated normalized action exactly matches a pending command. A changed action still supersedes the pending command, and the surviving command is revalidated at execution. The `queue_fix.py` overlay implements this correction without changing the original project source or raw runs.

In the archived correction, all 20 B0 D60 seeds were rerun through SUMO–BlueSky. The command issued at 360 s, and transport completed at 542 s, giving a duration of 242 s. Single-seed D00/D30 controls retain 182/212 s; fault times 359/361 s both give 242 s. B2's changed air-to-ground command retains a 302 s duration. B1 retains its air-to-air replacement sequence. Patched and unpatched control

reruns both give 342 s in that revision runtime, compared with the archived 341 s. This 1 s reproduction difference is unrelated to deduplication. We retain the archived cell because the correction does not change the command sequence. These regressions made no LLM calls. The original B4b traces contain genuine command replacement and remain unchanged.

Table S5 and Supplementary Figures S2 and S9 update only the affected B0 cell. D60 tests a specified event order; it does not establish a universal transport-delay threshold. Backend wall-clock time remains separate from controlled simulation waiting.

### S4.2 Physical transfer semantics and offline added-time sensitivity

The model replaces service assignments without tracking individual payloads. New air candidates route through the mission origin. Interrupted-service candidates route through V3 before the destination, but the supervisor can dispatch a backup before the original carrier reaches V3. Preemption interrupts and detaches the incumbent mission. Ground takeover starts a service timer based on the current reference corridor ETA. The model has no explicit states for unloading, rendezvous, loading, new inventory or re-shipment.

We evaluate added transfer time through an *offline trajectory-fixed sensitivity*, rather than a new closed-loop experiment. For an actually affected E2 mission with a post-failure DISPATCH, REASSIGN or GROUND_FALLBACK, we add one hypothesized transfer delay $h$ to recorded completion. The rule is the same for every policy; uninterrupted services receive no added time. A previously timely completion becomes late when $0 \leq d_m - c_m < h$. Choices, fault exposure and later scheduling remain fixed. The values $h = 15, 30, 60$ s are illustrative assumptions, not measured handling durations.

Table S16: Offline added transfer time: number of previously timely E2 replacements becoming late.

| Site | Policy | Replacements | Timely | +15 s | +30 s | +60 s |
|---|---|---|---|---|---|---|
| A | B1 | 200 | 80 | 0 | 0 | 0 |
| A | B2 | 200 | 80 | 0 | 0 | 0 |
| A | B4b | 191 | 75 | 1 | 1 | 2 |
| B | B1 | 200 | 80 | 0 | 80 | 80 |
| B | B2 | 200 | 80 | 0 | 80 | 80 |
| B | B4b | 199 | 80 | 0 | 80 | 80 |
| C | B1 | 200 | 0 | 0 | 0 | 0 |
| C | B2 | 200 | 0 | 0 | 0 | 0 |
| C | B4b | 201 | 0 | 0 | 0 | 0 |

At Site B, 80 affected replacements per coordinated policy are timely with 19 s remaining; adding 30 s makes all 80 late. At A, timely B1/B2 replacements have 60 s remaining, and none becomes late for $h \leq 60$ s. B4b has one such change at 15/30 s and two at 60 s. At C, the affected replacements in this subset are already late.

In the archived Site A preemption example, an extra 17 s of air-specific handling erases the realized speed advantage of 212 versus 229 s. More than 88 s also causes the emergency to miss its deadline. Adding the same handling time to both initial alternatives preserves their speed difference, so this threshold concerns an additional preemption step specifically. These checks identify conclusions that depend on omitted stages. They do not validate dynamic rescheduling with payload transfer.

### S4.3 Road-statistics boundary

The original morphology routine summed all imported passenger edges and divided by 10.24 $km^2$, without clipping them to its bbox argument. The revised calculation intersects each edge polyline with the configured geographic rectangle after projection through the SUMO network transform. It apportions SUMO edge lengths by the fraction of the polyline inside the rectangle and uses the same polygon's area for density. Intersection counts include only nodes inside the polygon, but count their distinct neighbors in the full graph to avoid artificial boundary intersections. Original full-network topology, OD samples and routing remain separately identified and unchanged. Site C still has higher intersection density than A; its description instead draws on dead ends, degree, circuity and facility distance.

### S4.4 Fault state machines and recovery conditions

Table S17: Six modeled fault families. Exact targets, geometry, times and windows are supplied in the YAML matrices.

| Fault | Trigger and state change | Independent local response | Service-recovery condition |
|---|---|---|---|
| F1 C2 loss | Target becomes non-commandable and in CONTINGENCY; mission interrupted | Return to V3 | Surviving compatible backup or ground; lost target remains excluded |
| F2 GNSS degradation | One-shot test of aircraft inside configured zone; GNSS DEGRADED | Occupied affected aircraft abandon mission and divert to V3 | Alternative with acceptable navigation and landing state; zone alone does not block other routes |
| F3 UTM change | DEGRADED blocks new noncritical air assignments; OUTAGE blocks all new air | OUTAGE interrupts in-flight missions and diverts them to V3 | Ground during outage; otherwise an eligible air assignment |
| F4 Landing closure | Configured site becomes unavailable; missions to it interrupted | Affected aircraft divert to V3 | Available compatible destination/continuation or ground |
| F5 Unknown aircraft | Prescribed noncooperative trajectory and temporary risk zone; affected missions interrupted | Commandable affected aircraft divert to V3 | Aircraft and route outside active zone, or ground; zone expires at configured end |
| F6 Flyaway | Target becomes uncontrolled and non-commandable; surrounding envelope can interrupt others | Target follows prescribed uncontrolled trajectory; other commandable affected aircraft divert | Exclude target and active envelope; use surviving air or ground |

The decision and execution checks evaluate whether zones apply within their configured active windows. Communication, navigation, UTM and landing-site states are not automatically repaired during these runs. Recovery means establishing a replacement service, not repairing the original aircraft or infrastructure.

## S5 Figure and data provenance

The compact manuscript figures are rebuilt for the narrow conference layout. Its Figure 1 is a vector diagram of the archived service model. Figure 2 is recalculated from the restored E1 per-run metrics with the same 20-seed block intervals. The extended figures below preserve the user-supplied ZIP assets. The archived conceptual illustration (Figure S4) contains AI-generated illustrative elements and editable diagram elements, as documented in the original research package; it depicts no observed deployment. Road and water geometry in Figure S5 comes from OpenStreetMap contributors. Facilities and air operations are modeled.

## S6 Restored repository and reproduction scope

The repository was recovered from commit `3078ffb60d58952b4c193b2edb1de7d85375bbab`, before the later research pivot. The current research branch is preserved. This paper and supplement use the independent branch `itsac-restored-20261016` at `https://github.com/AnXu-ITS/global-llm-mobility-reconfiguration`.

The restored `manuscript/source/revision_v3/all_metrics.json` contains 9,780 records for E1–E3, including 180 B4a ablation runs. `e4_run_metrics_revised.json` adds 1,360 E4 records, including the corrected B0 D60 cell. Together these represent 10,960 primary and 180 additional ablation runs. Configurations, prompts, schemas, simulation framework, revision scripts and 27 saved deterministic regression runs are also present. The supplied manuscript ZIP remains the authority for the original prose and figure assets.

The full historical `runs/` and `outputs/` directories were excluded from Git and are not recovered here. The old end-to-end postprocessing entry point depends on those missing inputs. Instead, `itsac/verify_restored_metrics.py` independently recomputes principal means, seed-block intervals, selected paired contrasts and corrected E4 timing from the committed per-run metrics. This is an offline verification; it does not rerun SUMO, BlueSky or an LLM. Detailed candidate logs and full raw simulation trajectories cannot be regenerated from aggregate metric records alone. See the supplement README for the file map, evidence limits and reproduction command.

## S7 Supporting transport outcomes and command sequence

Figures S1–S3 show failure recovery, the command sequence and paired compound-level contrasts. Figures S4–S9 retain the broader architecture, network maps, E1 outcomes and decisions, E3 loss and E4 sensitivity panels.

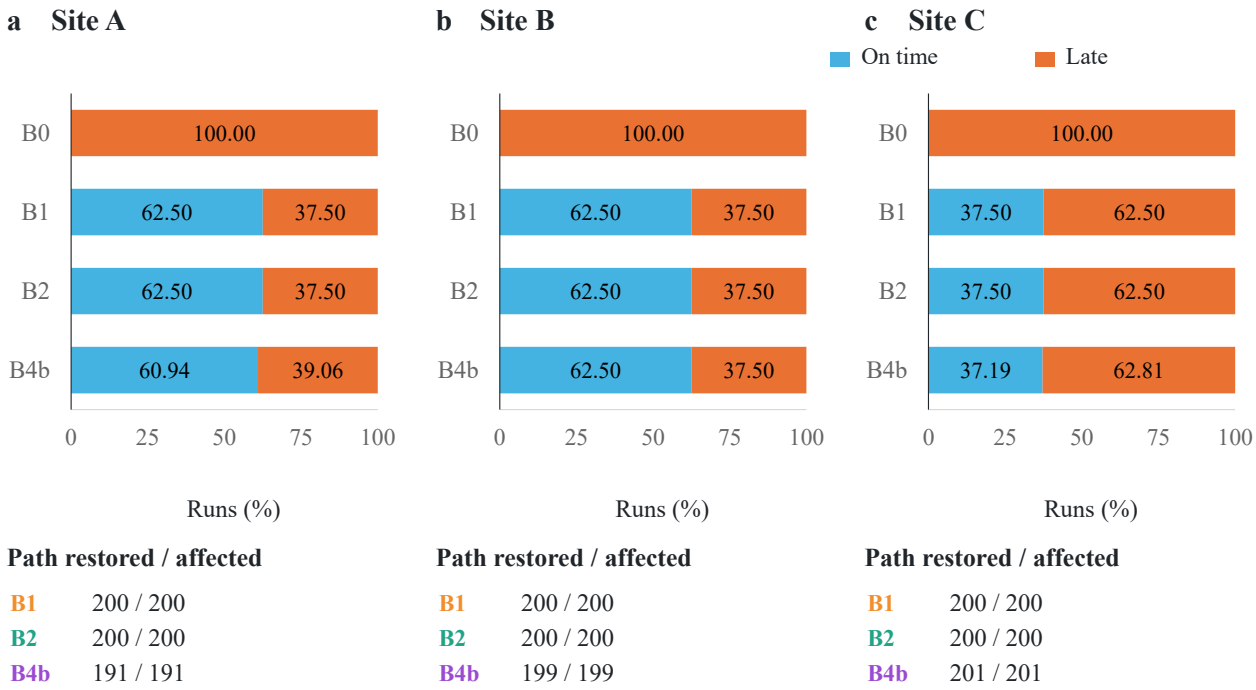



Figure S1: E2 path restoration and timely service answer different questions. (a–c) Bars show on-time and late completion percentages over all 320 runs per site and policy. The separate counts below give restored/actually affected primary air chains for B1/B2/B4b. Recovery denominators can depend on prior policy actions; B0 has no affected-air-chain denominator. Blue and orange encode deadline outcome here, not policy identity. Table S2 retains the full summaries and exact restoration counts.

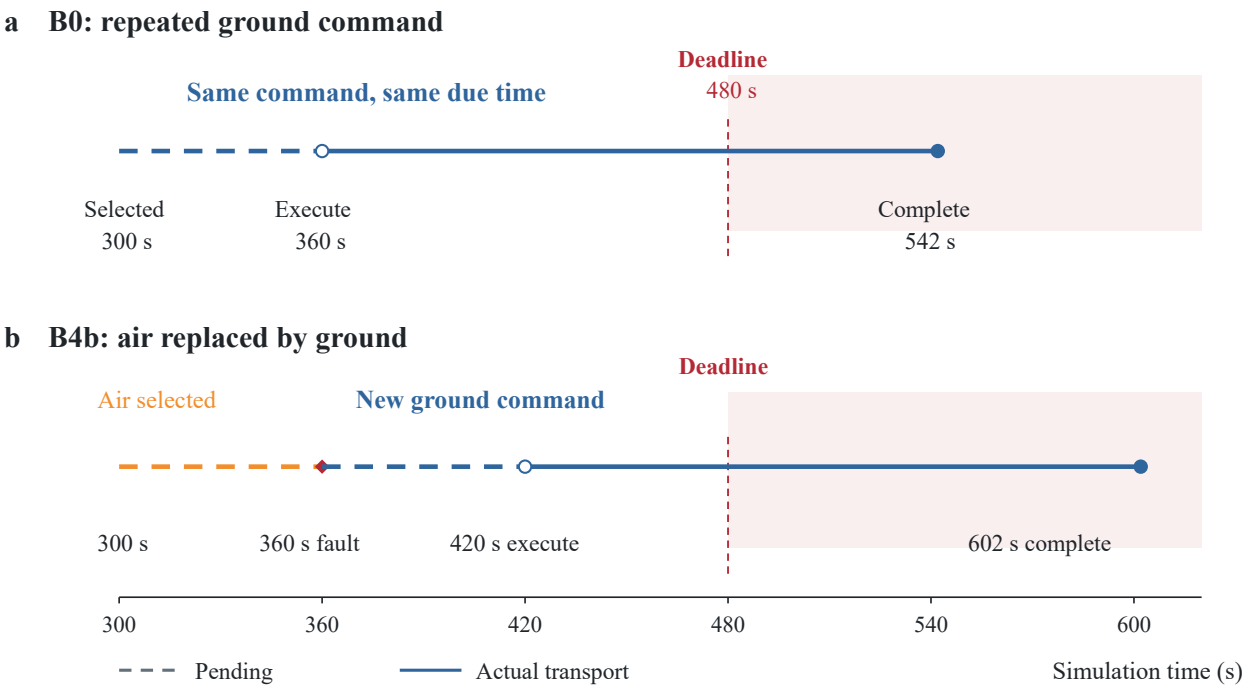



Figure S2: E4B D60 command sequences after identical-command deduplication. The pending air dispatch is superseded by a ground-fallback decision at the fault time. Ground executes at 420 s and completes at 602 s, after the 480 s deadline. Orange denotes the air command, blue the ground service and red the fault/deadline. Dashed lines denote pending commands and solid lines actual transport. Corrected B0 ground service executes at 360 s and completes at 542 s; B4b changes mode and completes at 602 s.




Figure S3: E3 paired coordination benefit by site and compound level. Points show mean B4b-minus-B0 differences in system weighted loss; whiskers are the original 95% seed-block intervals. Negative values indicate lower loss under B4b. Zero-width intervals coincide with their points. Each policy has 40 runs in L1/L4 and 80 in L2/L3. Levels jointly vary disturbance and demand. The corresponding policy–level means appear in Figure S8; exact values remain in Tables S3 and S7.

# S8 Additional network and policy comparisons

Table S18: Network and reference-service descriptors.

| Descriptor | A: Suzhou | B: Amsterdam | C: Edmonton |
|---|---|---|---|
| Statistical polygon area ($km^2$) | 10.2450 | 10.2361 | 10.2383 |
| Within-boundary road length (km) | 181.934 | 217.946 | 168.410 |
| Within-boundary road density ($km/km^2$) | 17.758 | 21.292 | 16.449 |
| Within-boundary intersection density ($km^{-2}$) | 37.970 | 100.331 | 39.655 |
| Full routing-network nodes | 542 | 1877 | 791 |
| Full directional passenger edges | 1286 | 3514 | 1681 |
| Full-network mean node degree | 2.830 | 2.663 | 2.420 |
| Full-network dead-end ratio | .1845 | .1193 | .2882 |
| Full-network sampled OD circuity | 1.1969 | 1.1409 | 1.4721 |
| D1–H1 network distance (km) | 2.9222 | 1.1192 | 3.4296 |
| D1–H1 Euclidean distance (km) | 1.7403 | .7246 | 2.5285 |
| D1–H1 circuity | 1.6791 | 1.5446 | 1.3564 |
| Reference closure ETA increase (%) | 37.87 | 88.69 | 34.76 |

Table S19: Paired E1 interface ablation: B4b minus B4a.

| Site | Endpoint | Pairs | Difference | 95% seed-block interval | Holm p |
|---|---|---|---|---|---|
| A | Completion (s) | 60 | -13.567 | [-15.978, -11.155] | 9.33e-07 |
| A | Damaged services/run | 60 | -0.383 | [-0.441, -0.326] | 3.27e-07 |
| A | Air intervention (pp) | 60 | -38.333 | [-44.093, -32.574] | 0.001953 |
| B | Completion (s) | 60 | 1.450 | [-2.288, 5.188] | 1.000 |
| B | Damaged services/run | 60 | -0.017 | [-0.054, 0.021] | 1.000 |
| B | Air intervention (pp) | 60 | -1.667 | [-5.437, 2.104] | 1.000 |
| C | Completion (s) | 60 | 0.000 | [0.000, 0.000] | 1.000 |
| C | Damaged services/run | 60 | 0.000 | [0.000, 0.000] | 1.000 |
| C | Air intervention (pp) | 60 | 0.000 | [0.000, 0.000] | 1.000 |

Table S20: Selected E2 paired comparisons: B4b minus B2.

| Site | Endpoint | Pairs | Difference | 95% seed-block interval | Holm p |
|---|---|---|---|---|---|
| A | Completion (s) | 320 | 4.525 | [2.412, 6.638] | 0.003069 |
| A | Recovery time (s) | 191 | 3.927 | [0.930, 6.924] | 0.142465 |
| B | Completion (s) | 320 | 0.175 | [0.060, 0.290] | 0.055710 |
| B | Recovery time (s) | 199 | 0.281 | [0.098, 0.465] | 0.055710 |
| C | Completion (s) | 320 | 0.922 | [-0.214, 2.057] | 1.000 |
| C | Recovery time (s) | 200 | 0.255 | [-0.138, 0.648] | 1.000 |

Network densities use clipped study polygons; full-network topology and sampled OD circuity retain the imported routing network. The interface comparison has 60 matched scenario–seed pairs per site, aggregated over ten seed blocks. Each site retains a three-endpoint Holm family. The E2 completion comparison uses all 320 pairs; recovery time uses only jointly affected pairs and retains the original 12-endpoint Holm family. These tables correspond to the detailed results summarized in the eight-page paper.

# S9 Extended figure panels

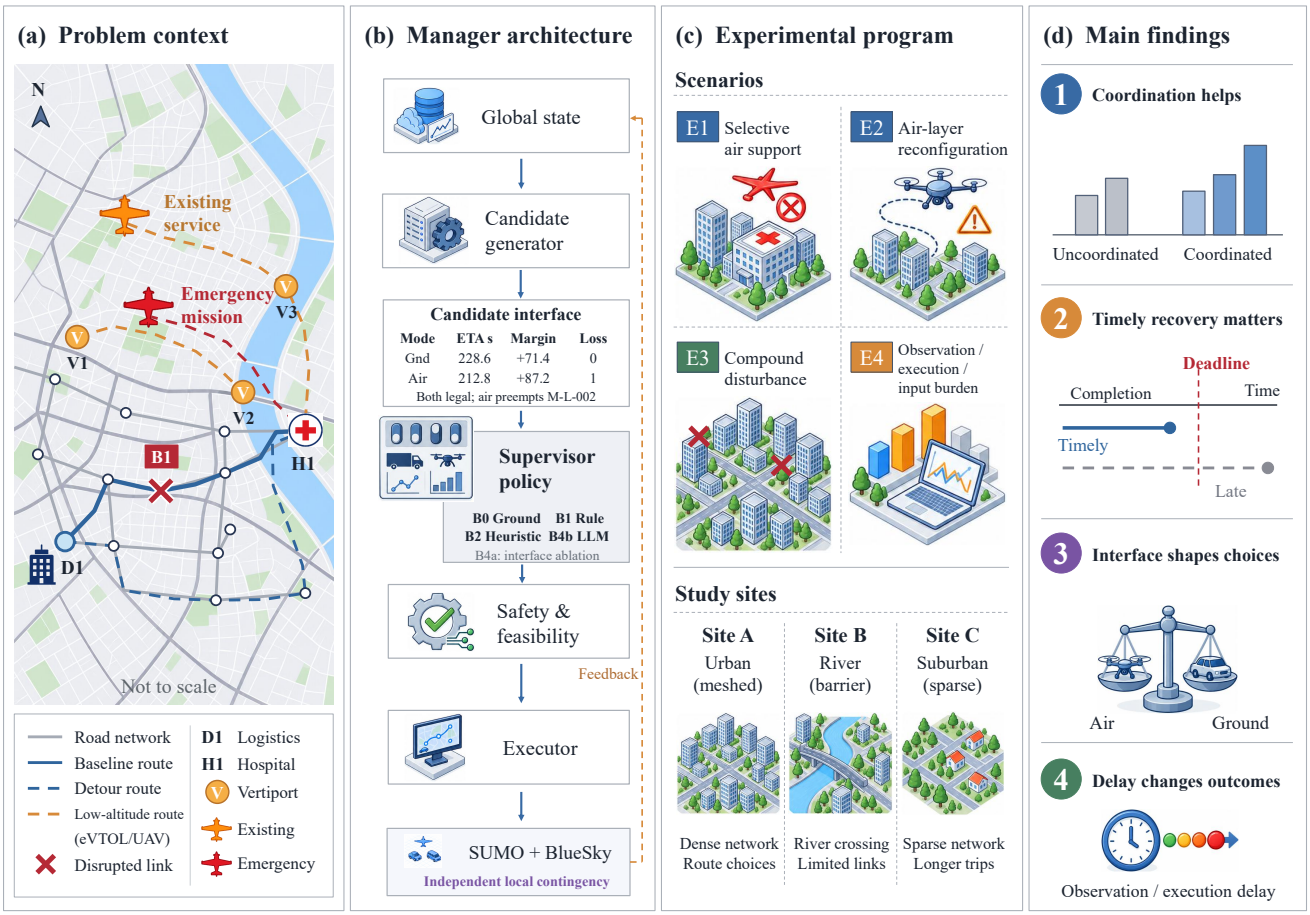



Figure S4: Archived conceptual overview of the service system, candidate interface and experiment panels. The map and finding graphics are qualitative illustrations, not measured outcomes. B0/B1/B2/B4b are the primary policies; B4a is the interface ablation.

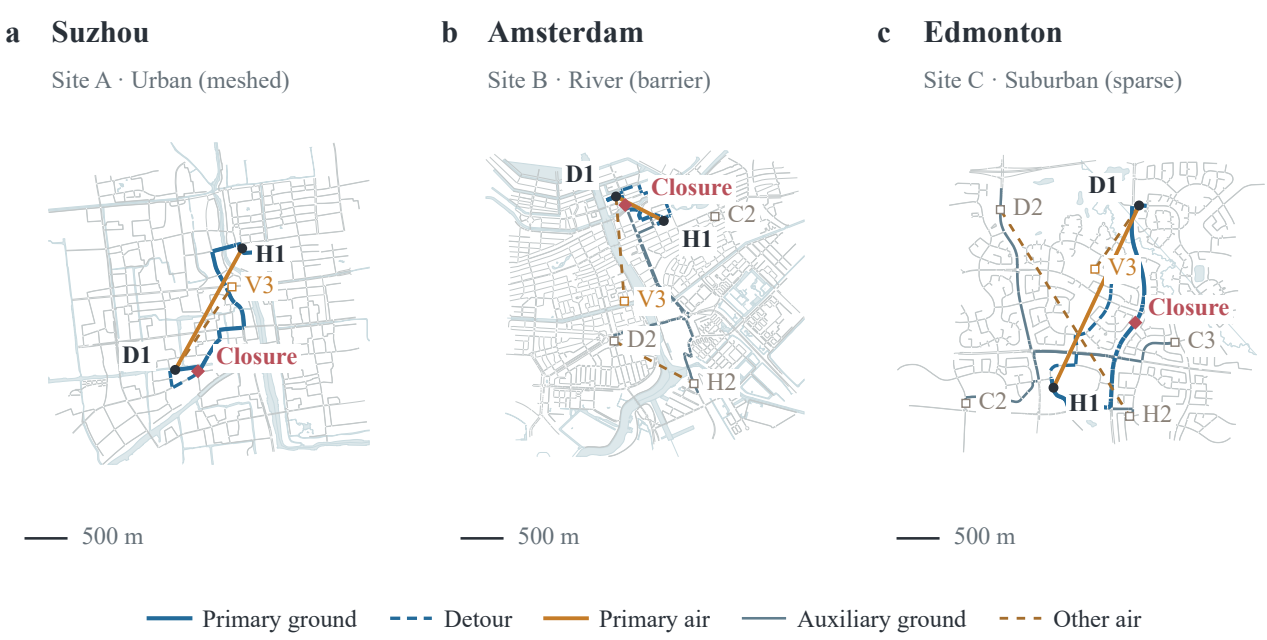



Figure S5: Study networks in Suzhou (A), Amsterdam (B) and Edmonton (C), using a common projected scale. Blue marks ground connections and orange marks modeled air connections. D1/H1 are reference service endpoints and V3 is a backup landing site. The full imported networks can extend outside the nominal study rectangles. Road and water data: OpenStreetMap contributors; services and disruptions are modeled.

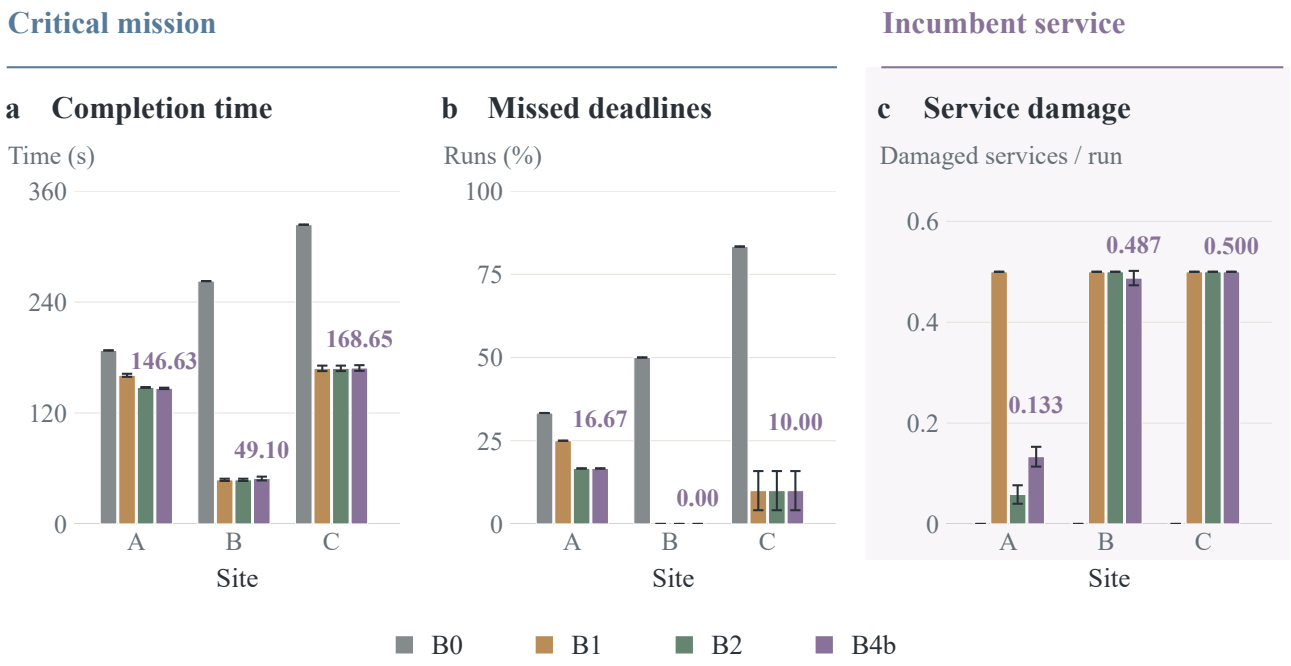
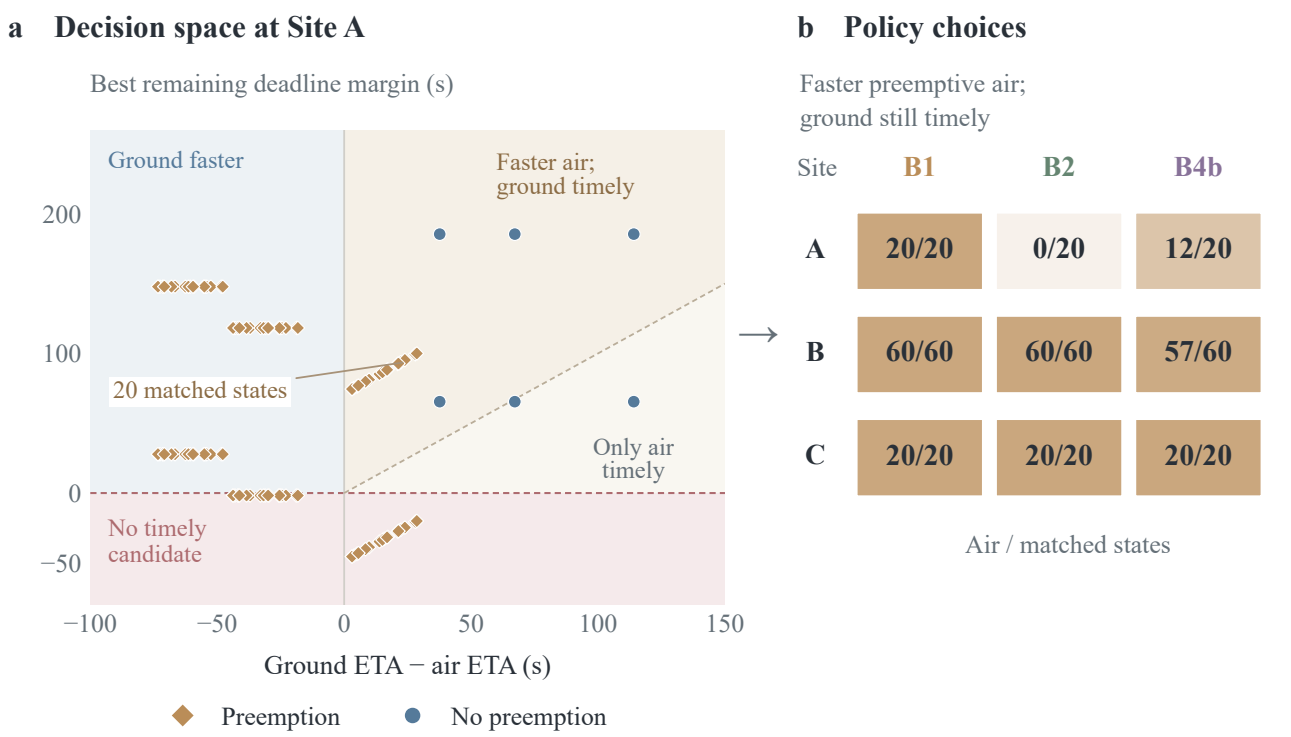



Figure S6: Extended E1 result panel from the supplied manuscript: mean completion time, missed deadlines and final-state incumbent damage. Each site–policy cell contains 240 runs and 20 seed blocks. Error bars retain the 95 percent seed-block intervals.




| c **No candidate predicted timely** | Site A | Site B | Site C |
|---|---|---|---|
| Observed late / states, per policy | **40/40** | No states | **24/27** |

Figure S7: E1 decision regimes. The candidate geometry separates timing advantage from preemption. Faster air requires preemption while ground remains timely in 20, 60 and 20 matched states at A, B and C. All first candidate snapshots match across B1/B2/B4b in the 720 scenario–seed states. Candidate margins are approximate, not guaranteed outcomes.

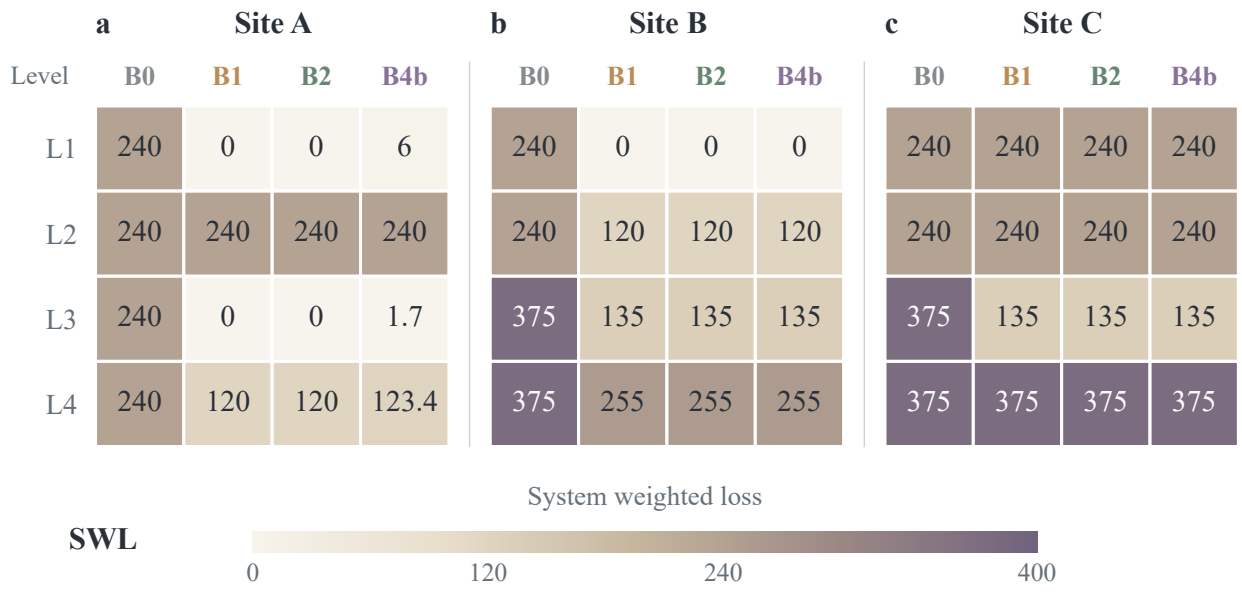



Figure S8: E3 weighted service loss by compound disturbance and task level. Each policy has 40 runs in L1/L4 and 80 in L2/L3. The levels change both failure structure and task demand. Exact means are retained in Tables S3 and S7; paired contrasts appear in Figure S3.

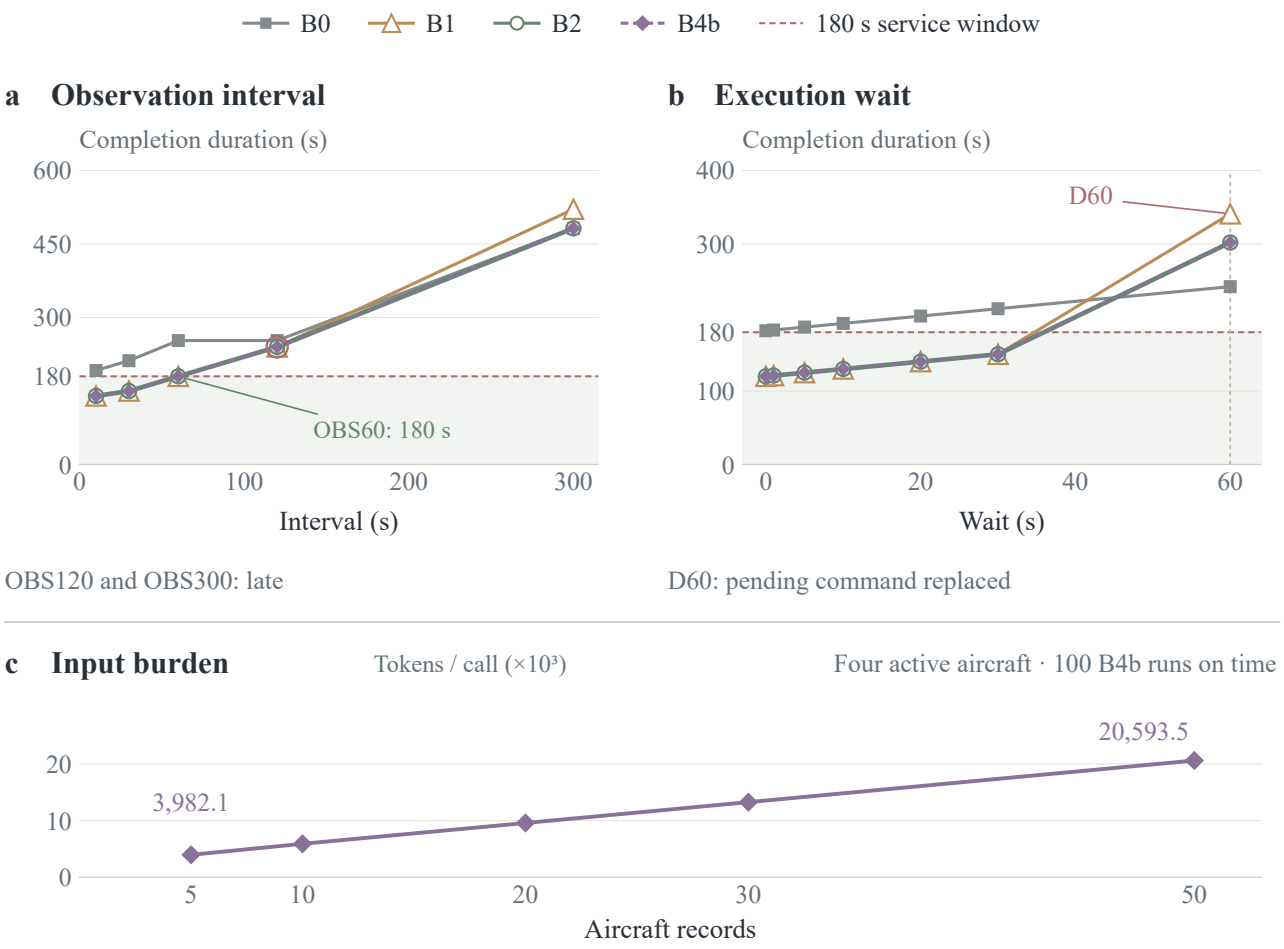



Figure S9: E4 observation, execution waiting and input burden at Site A. Each policy–arm completion result has 20 runs. The dashed threshold is the 180 s service window. Input records increase from 5 to 50 around an unchanged four-aircraft core fleet; these are not larger active fleets.